

EV Data Analytics & Visualization Dashboard

Project Description:

The **EV Data Analytics & Visualization Dashboard** is a Tableau-based business intelligence solution developed to analyze Electric Vehicle (EV) charging patterns, battery performance, and vehicle efficiency. The system collects EV datasets from CSV files or a MySQL database, preprocesses the data using Python, and presents meaningful insights through interactive Tableau dashboards.

The project enables users to identify charging trends, monitor battery health, compare EV models, and analyze energy consumption. It supports city planners, fleet managers, and EV buyers in making informed decisions using visual analytics.

Built using **Python, Flask, MySQL, Tableau Desktop, Tableau Public, HTML, CSS, and Bootstrap**, the solution provides an easy-to-use web interface with embedded Tableau dashboards.

Scenarios

Scenario 1

A **City Energy Planner** uploads EV charging data into the system. The dashboard analyzes charging time, duration, and electricity consumption to identify peak charging hours and high-demand locations. These insights help optimize charging infrastructure and maintain grid stability.

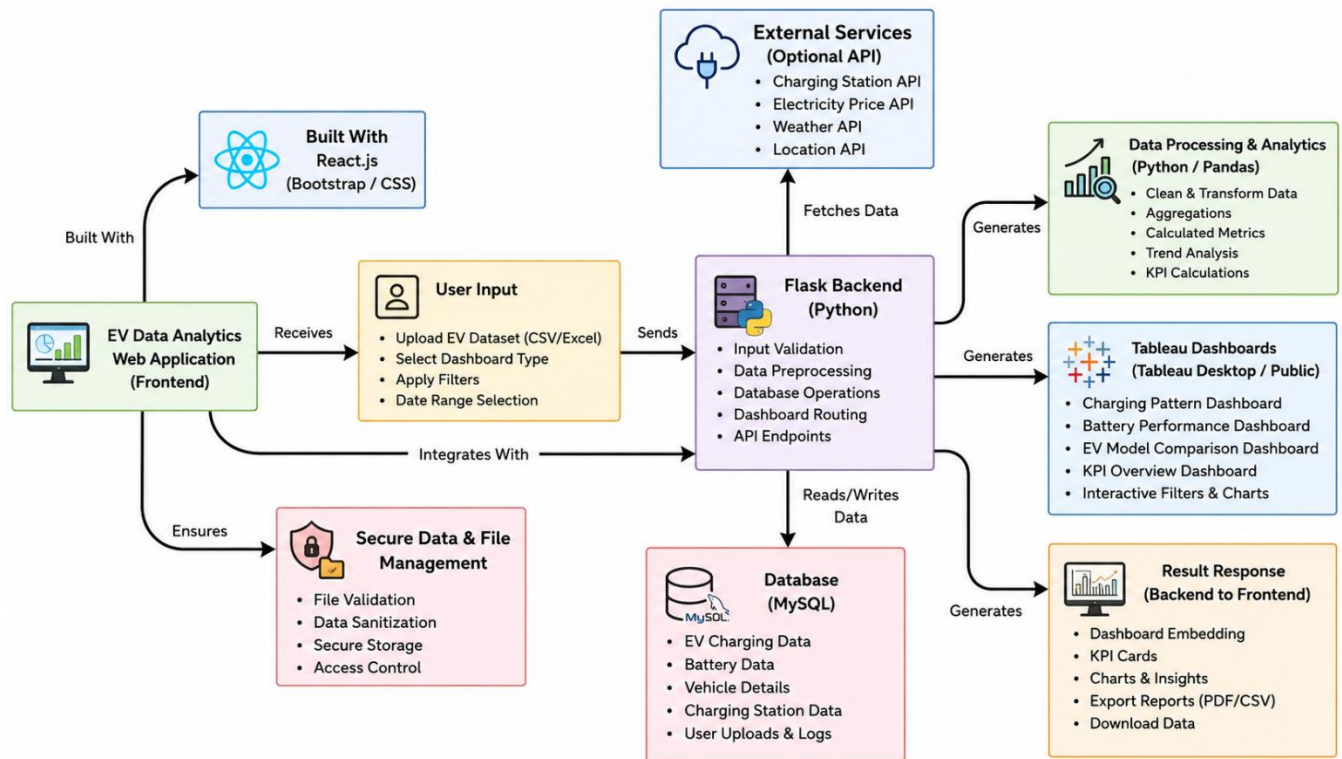
Scenario 2

An **EV Fleet Manager** monitors battery health, charging cycles, energy efficiency, and driving range using interactive dashboards. The system compares predicted and actual driving range, helping identify batteries that require maintenance or replacement.

Scenario 3

An **EV Buyer** compares different electric vehicle models based on driving range, charging cost, battery capacity, efficiency, and environmental impact. Interactive filters allow personalized comparisons, helping users choose the most suitable EV.

Technical Architecture:



Description: The EV Data Analytics & Visualization Dashboard follows a modular three-tier architecture. The frontend is developed using HTML, CSS, and Bootstrap, while the backend uses Flask to manage data uploads, database connectivity, and dashboard embedding. Python performs data preprocessing, and MySQL stores EV datasets. Tableau connects to the database to generate interactive dashboards and reports. The complete application can be deployed locally or hosted on Render, with Tableau dashboards published on Tableau Public.

Pre-requisites:

- Python Programming
- Flask Framework
- Pandas
- NumPy
- MySQL
- Tableau Desktop
- Tableau Public
- HTML
- CSS
- Bootstrap
- VS Code
- Git & GitHub

Project Workflow:

Milestone 1

Dataset Collection & Architecture

Activities

- Collect EV Dataset
- Design Solution Architecture
- Setup MySQL Database
- Configure Development Environment

Milestone 2

Core Functionalities

Activities

- Dashboard Development
- Dataset Upload
- Database Integration
- Data Validation

Milestone 3

Backend Development

Activities

- Flask API
Development
- Data Cleaning
- Data
Preprocessing
- MySQL
Integration

Milestone 4

Tableau Dashboard

Development

Activities

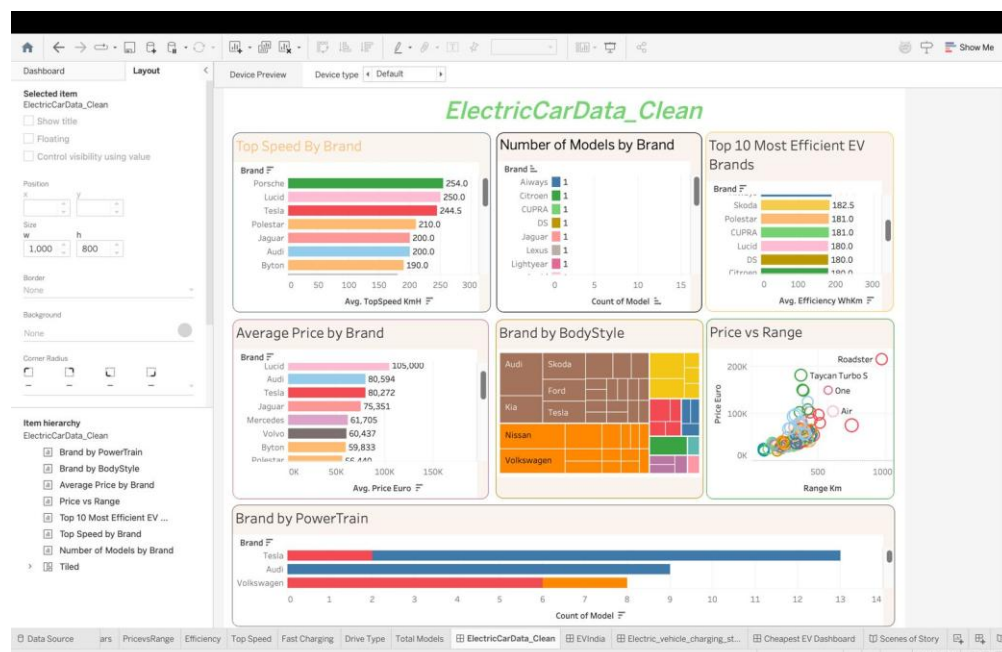
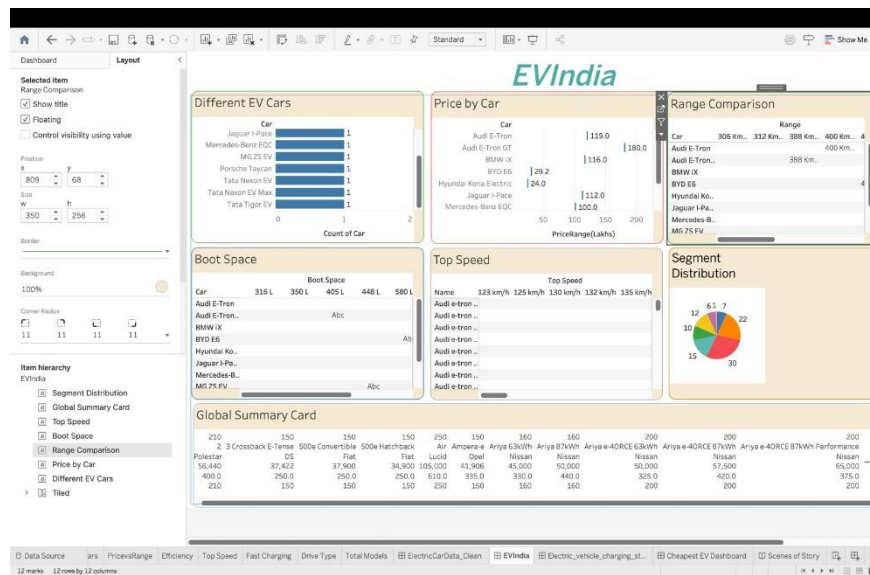
- Charging
Dashboard
- Battery
Dashboard
- EV
Comparison
Dashboard
- KPI Dashboard
- Story
Dashboard

Milestone 5

Deployment

Activities

- Deployment
- GitHub Push
- Publish
- Tableau
Dashboard
- Deploy Flask
Application



Conclusion:

The **EV Data Analytics & Visualization Dashboard** is a business intelligence solution developed using **Python, Flask, MySQL, Tableau Desktop, and Tableau Public** to help city energy planners, fleet managers, and EV consumers analyze electric vehicle data effectively. By analyzing **charging patterns, battery performance, driving range, energy consumption, charging costs, and vehicle efficiency**, the system provides interactive dashboards, KPI metrics, trend analysis, and comparative insights that support informed decision-making.

The project demonstrates how **data analytics and visualization** can improve charging infrastructure planning, optimize fleet management, reduce operational costs, and promote sustainable transportation. Through interactive Tableau dashboards, users can identify peak charging hours, monitor battery health, compare EV models, and evaluate long-term performance, making EV adoption more efficient and data-driven.

Looking ahead, the project offers significant opportunities for enhancement, including **real-time EV charging station integration, IoT-enabled battery monitoring, AI-based demand forecasting, predictive battery health analysis, live dashboard refresh, cloud-based analytics, mobile application support, and smart city integration**. By continuously improving the analytics platform and expanding its features, the **EV Data Analytics & Visualization Dashboard** has the potential to become a comprehensive EV decision-support system that benefits government agencies, fleet operators, charging station providers, researchers, and EV users while supporting the growth of sustainable and intelligent transportation systems.